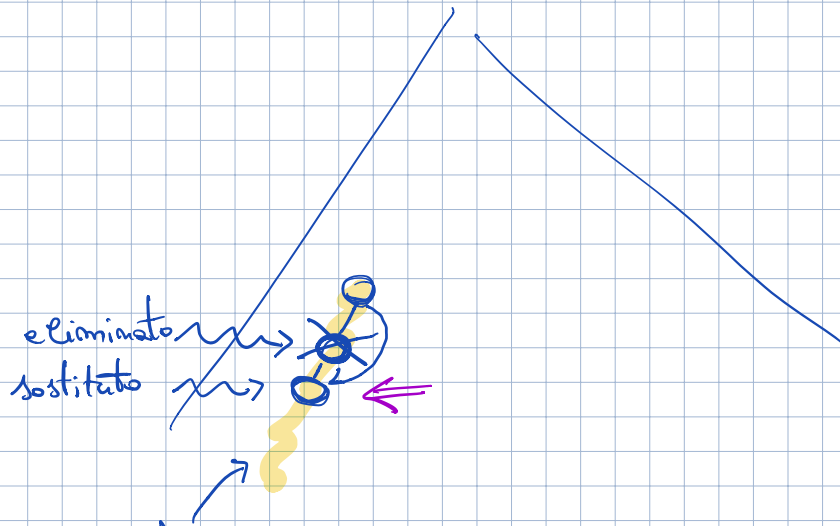


Cancellazione RBT

Se è stato eliminato un nodo BLACK
potrei aver creato un problema



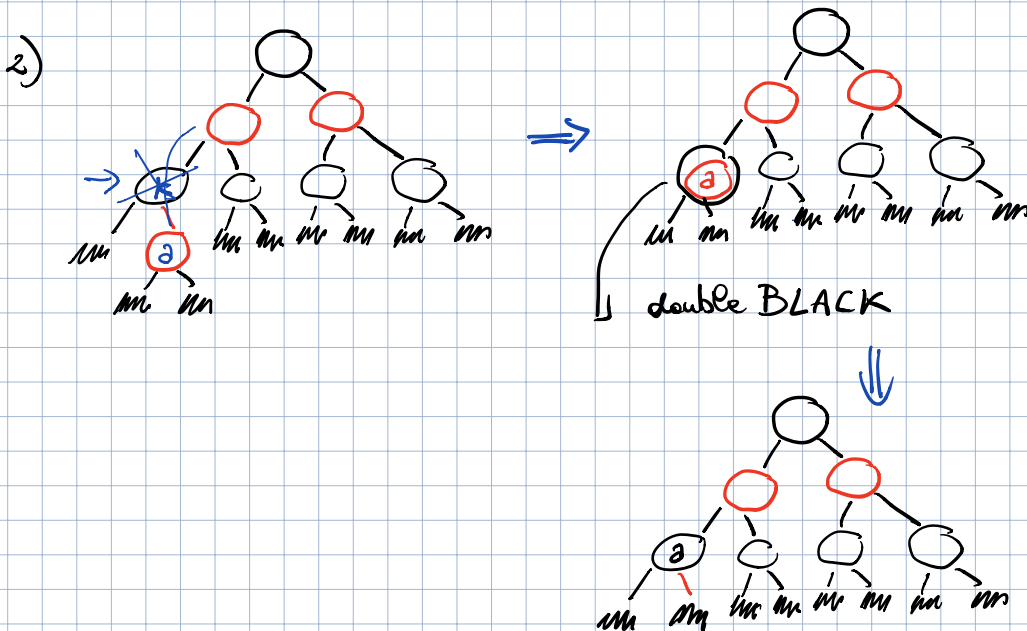
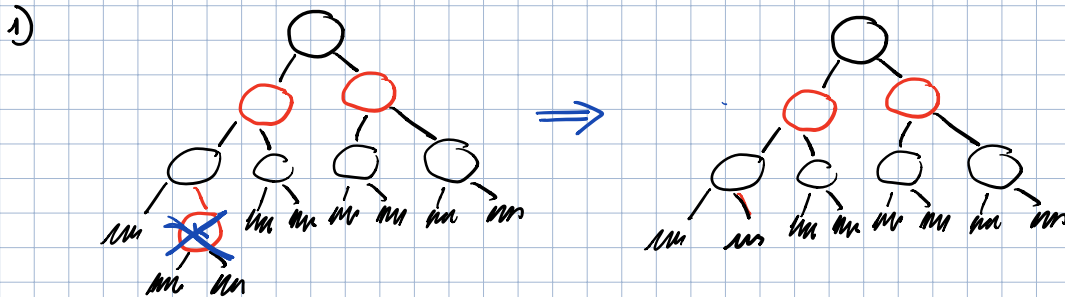
§

TRUCCO il nodo
che sostituisce quello
eliminato conta temporaneamente
come 2 nodi BLACK → double-BLACK

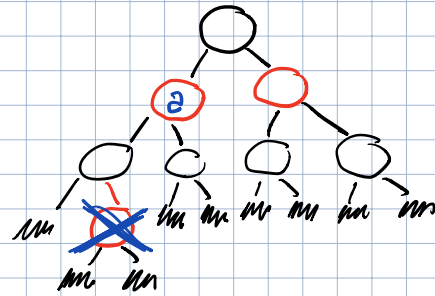
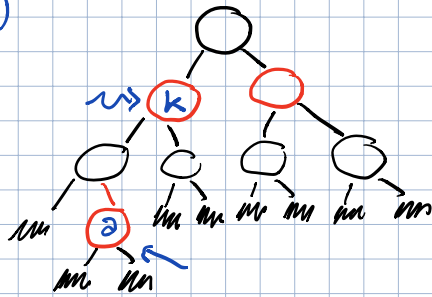
Se il nodo double-BLACK era prima RED
lo ricoloro BLACK e termino

Altimenti applico ROTAZIONI e RICOLOR.
per cercare di risolvere o spostare verso
C'alto il problema

Example

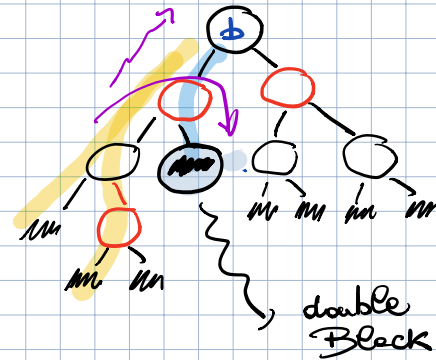
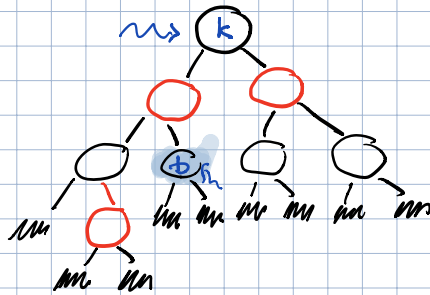


3)



Usando predecessore

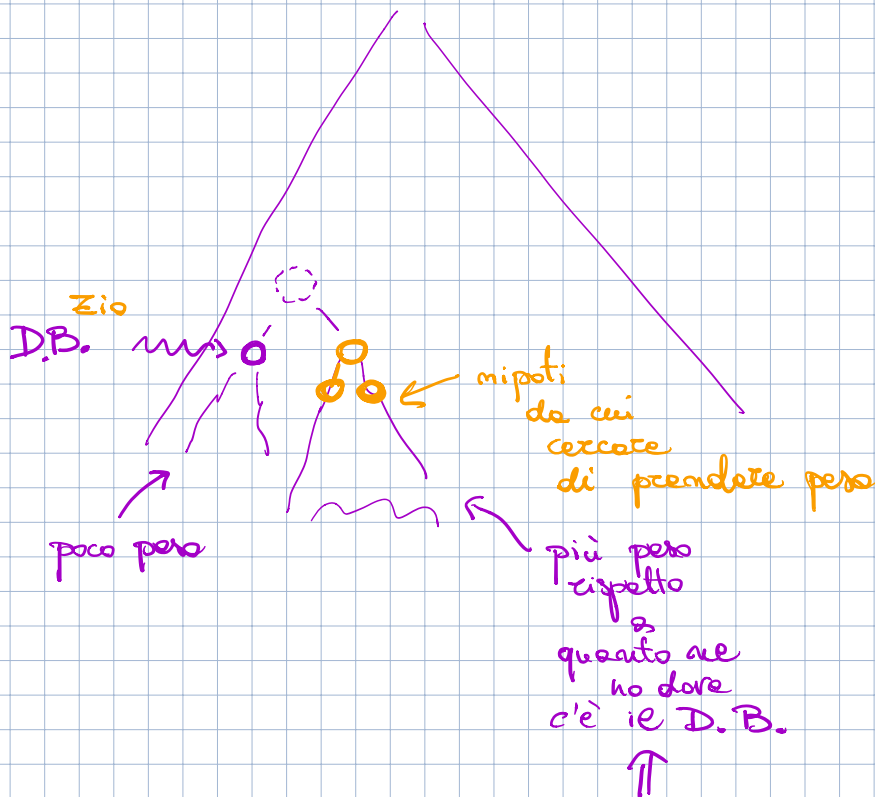
4)



Problema
da risolvere
con Rot. e Ric.

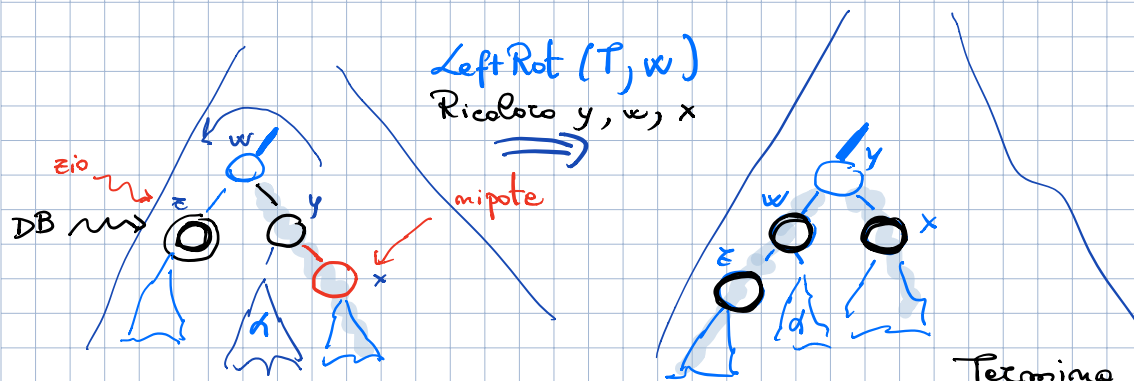
Rot. e Ric per la Concezione

4 CASI



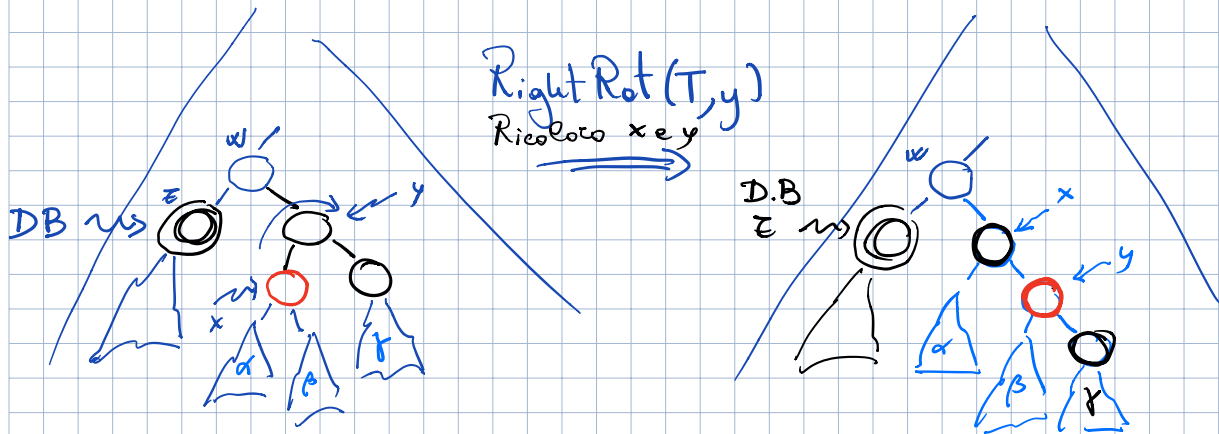
1) Caso Fortunato

Il D.B. ha un NIPOTE RED opposto
a Cui



2) Caso Quasi Fortunato

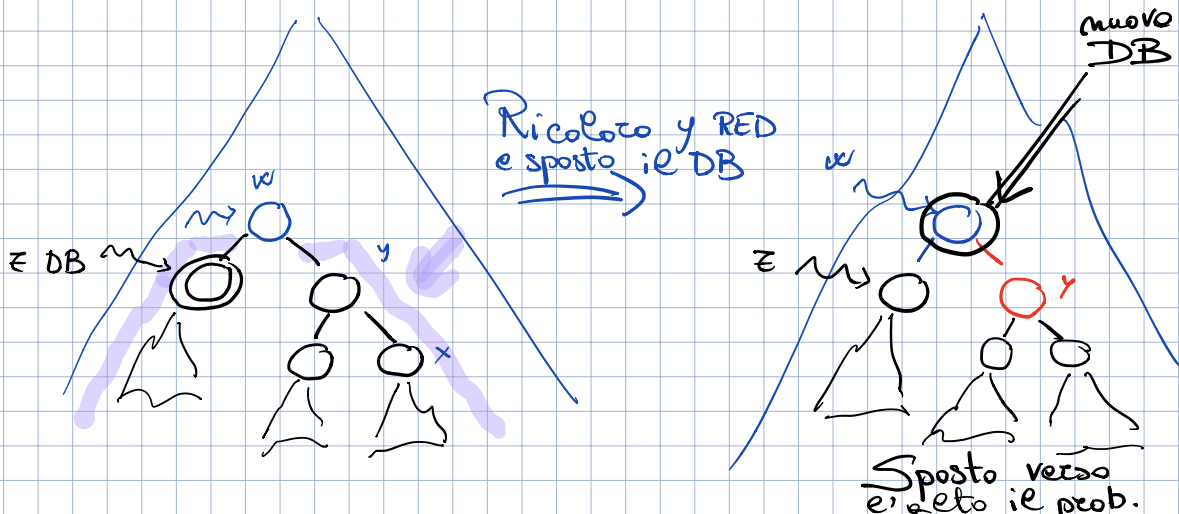
Il D.B. ha un nipote RED "dello stesso lato"



Poi applico il caso fortunato e termino

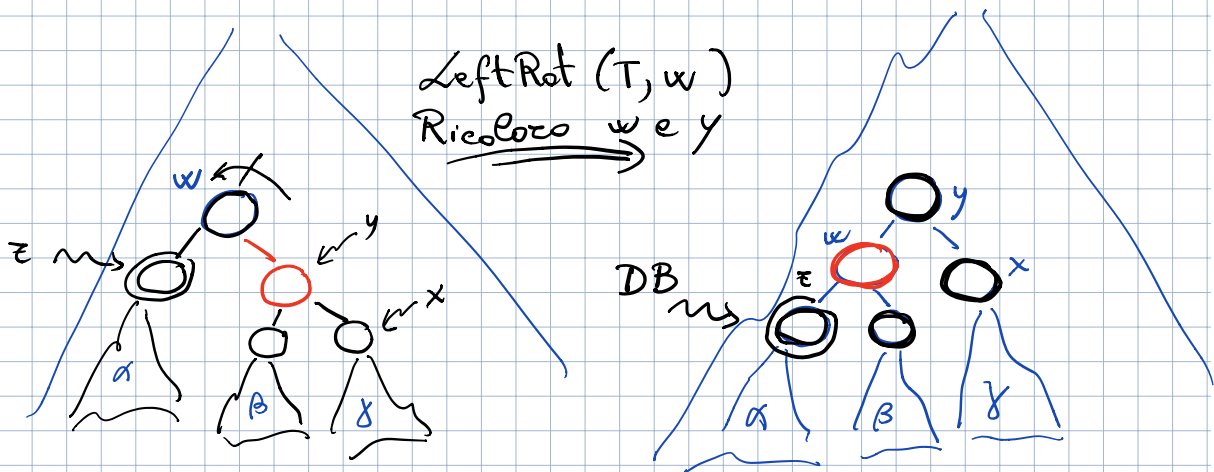
3) Caso Sfortunato

Entrambi i nipoti e il fratello del DB sono BLACK



4) Caso Antipatico

Entrambi i nipoti del DB sono BLACK
 Il fratello del DB è RED

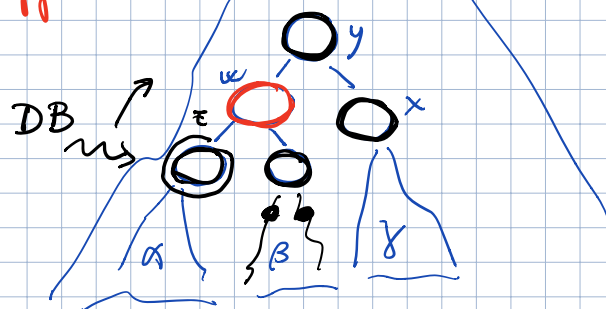


Ac
 passo
 successivo

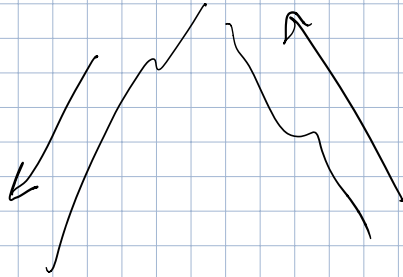
Caso 1) oppure
 Caso 2) oppure

Caso 3)

ma se
 sono nel
 caso 3) sposto
 il problema
 su
 $w \rightarrow$ si ricoloca
 e termino



Completamento



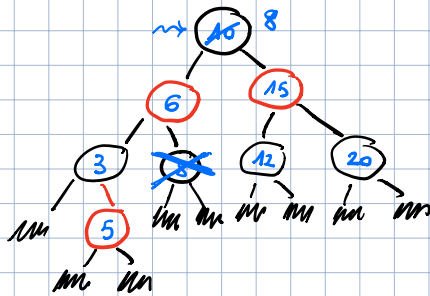
$$\Theta(\log n) + O(\log n)$$

||

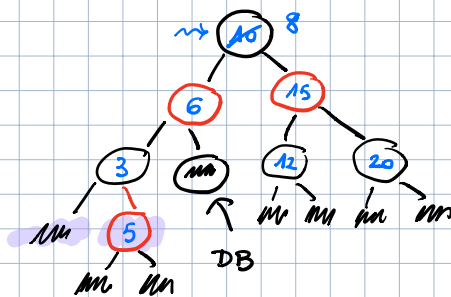
$$\Theta(\log n)$$

Esempio

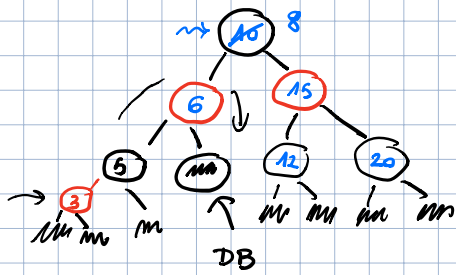
Predecessore



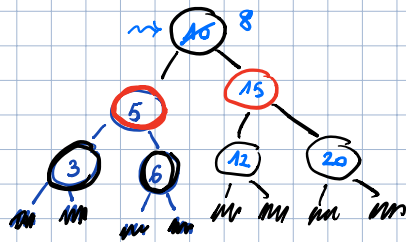
Cancellare 10



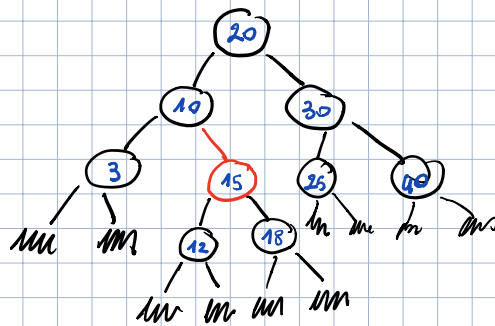
Case 2



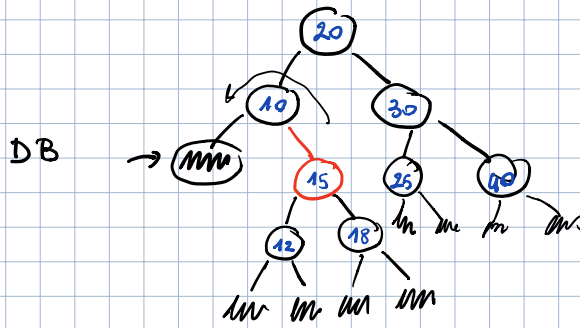
Case 1

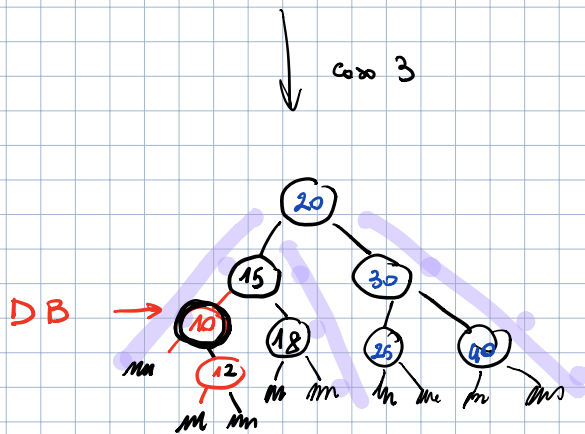
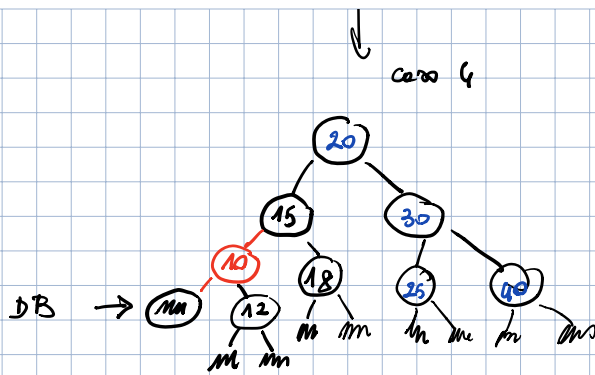


Esempio



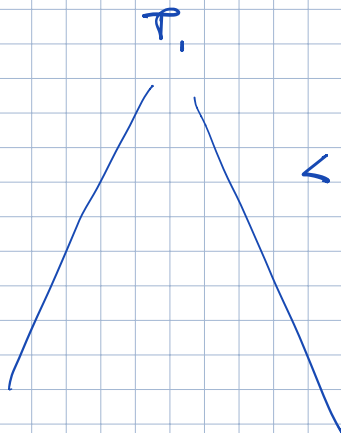
Cancelare 3



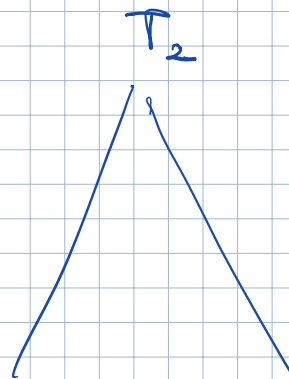


~ . ~ . ~ .

RBT Join

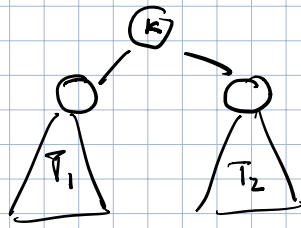


< (K) <

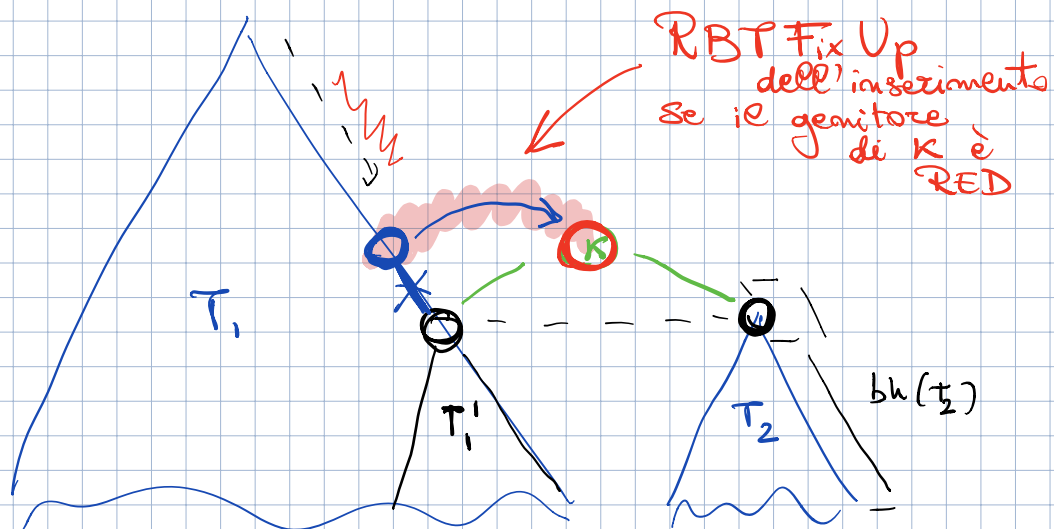


T_1, T_2 RBT

$$\text{lh}(\tau_1) = \text{lh}(\tau_2)$$



$$\text{bh}(\tau_1) > \text{bh}(\tau_2)$$



$$O(\max(\text{bh}(T_1), \text{bh}(T_2)))$$

$$O(\log n) + O(\log n) \rightarrow O(\log n)$$

Che complessità avrei avuto inserendo in T_1 tutte le chiavi di T_2 e K ?